

DATA ANALYTICS PROJECT REPORT

E-COMMERCE BUSINESS ANALYTICS

An End-to-End Data Analytics Project Using Excel, MySQL, Python and Power BI

Project Type	Business & Data Analytics
Dataset	Sample Superstore
Tools	Excel, MySQL, Python, Pandas, Matplotlib, Power BI, DAX
Analysis Period	2015 - 2018

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1. Project Overview

The E-Commerce Business Analytics project is an end-to-end data analytics project focused on understanding sales performance, profitability, customer behavior, product performance, regional trends, and the relationship between discounts and profitability.

The project uses the Sample Superstore transactional dataset and follows a complete analytics workflow, beginning with data cleaning and validation and progressing through SQL-based business analysis, Python exploratory analysis, and interactive Power BI visualization.

The primary objective is to transform raw transactional data into meaningful business insights that can help identify strong-performing areas, loss-making areas, profitability patterns, and potential opportunities for business improvement.

The project demonstrates the complete analytical process:



2. Project Objectives

The main objectives of this project are:

1. Analyze overall business revenue and profitability.
2. Understand category and sub-category performance.
3. Identify products and sub-categories contributing to losses.
4. Analyze regional sales and profitability differences.
5. Evaluate customer revenue and profitability.
6. Examine business performance over time.
7. Analyze the relationship between discount levels and profitability.
8. Identify high-performing and low-performing business areas.
9. Develop an interactive Power BI dashboard for business reporting.
10. Convert analytical findings into clear business insights and recommendations.

3. Business Questions

The analysis was designed to answer the following business questions:

1. How is the overall business performing in terms of revenue and profit?
2. Which product categories and sub-categories generate the most revenue and profit?
3. Which products or sub-categories are contributing to losses?
4. How does profitability vary across different regions?
5. How has revenue and profit changed over time?
6. How do different discount levels relate to profitability?
7. Which customers generate the highest revenue and profit?
8. Does high sales revenue always translate into high profitability?
9. Which areas of the business demonstrate strong profitability?
10. Where are the key opportunities for improving business performance?

4. Dataset Description

The project uses the Sample Superstore dataset, which contains transactional e-commerce sales information covering customers, orders, products, categories, regions, sales, discounts, quantities, and profit.

The cleaned dataset contains:

Dataset Attribute	Value
Records	9,994
Columns	21
Duplicate Records	0
Missing Postal Code Values	11
Analysis Period	2015–2018

The dataset contains information that allows the business to be analyzed from multiple perspectives, including:

Orders	Customers	Products
Categories	Sub-categories	Regions
States	Segments	Sales
Quantity	Discounts	Profit
Order dates	Ship dates	Ship modes

5. Data Preparation and Cleaning

Before performing business analysis, the dataset was cleaned and validated to improve data quality and consistency.

The data preparation process included:

5.1 Duplicate Removal

Duplicate records were identified and removed during the cleaning process.

RESULT

The final cleaned dataset contains 9,994 records with 0 duplicate records.

5.2 Missing Value Analysis

Missing values were reviewed across the dataset.

RESULT

The final cleaned dataset contains 11 missing Postal Code values.

These missing values were documented and retained where appropriate rather than introducing unsupported values.

5.3 Data Type Standardization

Data types were reviewed and standardized for analysis.

Field Type	Fields	Standardized To
Numerical	Sales, Discount, Profit, Quantity	Appropriate numeric formats
Date	Order Date, Ship Date	Proper date formats for time-based analysis

5.4 Data Validation

The cleaned dataset was validated by checking:

Dataset dimensions	Data types	Missing values
Duplicate records	Numerical summaries	Date fields
Sales values	Discount values	Profit values

The cleaned data was then prepared for MySQL, Python, and Power BI analysis.

6. Analytical Workflow

The project follows a structured end-to-end analytics workflow.

Step	Stage	Description
1	Excel	Excel was used for initial data cleaning and validation.
2	MySQL	The cleaned dataset was loaded into MySQL and analyzed using business-focused SQL queries.
3	Python	Python was used for data validation, exploratory data analysis, business analysis, data aggregation, and pattern identification.
4	Power BI	Power BI was used to transform the analytical results into an interactive business dashboard.
5	Business Insights	The analytical results were interpreted to identify important business patterns and potential improvement areas.
6	Documentation	The final findings were documented through the project README and this project report.

7. Tools and Technologies

Tool / Technology	Purpose
Excel	Initial data cleaning and validation
MySQL	Data storage and SQL business analysis
Python	Data cleaning, EDA, and business analysis
Pandas	Data manipulation and transformation
Matplotlib	Exploratory data visualization
Power BI	Interactive dashboard and reporting
DAX	KPI and analytical measure calculations
GitHub	Version control and project documentation
GitHub Desktop	Git repository workflow management

8. MySQL Analysis

MySQL was used to store the cleaned transactional dataset and perform business-focused SQL analysis.

AT A GLANCE

A total of 16 SQL business queries were developed to investigate different aspects of business performance.

The analysis covered:

Overall business KPIs	Revenue and profit	Category performance
Sub-category performance	Regional performance	Customer performance
Product performance	Discount impact	Year-over-year performance
Ship mode analysis	State-level performance	

9. Key Business Metrics

The overall business performance calculated from the dataset is:

Metric	Result
Total Revenue	\$2,297,201.07
Total Profit	\$286,397.79
Profit Margin	12.47%
Total Orders	5,009
Total Customers	793
Total Units Sold	37,873
Average Order Value	\$458.61

These metrics provide the baseline for evaluating category, regional, product, customer, and discount-level performance.

10. Category Performance Analysis

The analysis identified significant differences in profitability across the three major product categories.

Category	Revenue	Profit	Profit Margin
Technology	\$836,154.10	\$145,455.66	17.40%
Furniture	\$741,999.98	\$18,451.25	2.49%
Office Supplies	\$719,046.99	\$122,490.88	17.04%

Technology generated the highest revenue among the three categories and also produced a strong profit margin.

Office Supplies also demonstrated strong profitability relative to the overall business.

Furniture generated substantial revenue but produced a significantly lower profit margin.

KEY OBSERVATION

This indicates that high revenue does not necessarily translate into high profitability.

11. Sub-Category Profitability Analysis

Sub-category analysis revealed significant differences in profitability. Some sub-categories generated strong positive margins, while others contributed to losses.

Strong-Performing Sub-Categories

Sub-Category	Profit	Profit Margin
Copiers	\$55,617.90	37.20%
Paper	\$34,053.34	43.39%

These results indicate strong profitability within these product areas.

Loss-Making Sub-Categories

Sub-Category	Profit	Profit Margin
Tables	-\$17,725.59	-8.56%
Bookcases	-\$3,472.56	-3.02%
Supplies	-\$1,188.99	-2.55%

AREA FOR INVESTIGATION

Tables represent a particularly important area for investigation because of their negative profitability despite contributing to overall Furniture revenue.

12. Regional Analysis

Regional analysis showed differences in revenue and profitability.

Region	Revenue	Profit	Profit Margin
West	\$725,457.93	\$108,418.79	14.94%
East	\$678,781.36	\$91,522.84	13.48%
Central	\$501,239.88	\$39,706.45	7.92%
South	\$391,721.90	\$46,749.71	11.93%

The West region generated the highest revenue among the four regions and had a 14.94% profit margin.

The Central region had the lowest profit margin at 7.92%.

This demonstrates that business performance varies not only by sales volume but also by regional profitability.

13. Discount and Profitability Analysis

Discount analysis was performed to investigate how different discount levels were associated with aggregated profitability.

The analysis showed the following profit values at different discount levels:

Discount	Aggregated Profit
0%	\$320,987.88
20%	\$90,338.16
30%	-\$10,369.34
40%	-\$23,057.08
50%	-\$20,506.51
60%	-\$5,944.64
70%	-\$40,075.46
80%	-\$30,539.13

Aggregated profit was negative at the 30% discount level, with several higher discount levels also showing losses.

This finding highlights the importance of monitoring discount strategies together with revenue and profitability rather than evaluating discounts based on sales volume alone.

INTERPRETATION NOTE

The analysis describes an observed relationship in this dataset; it does not by itself establish that discounts are the sole cause of the observed losses.

14. Year-over-Year Performance

The analysis also examined business performance from 2015 to 2018.

Year	Revenue	Profit
2015	\$484,247.56	\$49,544.06
2016	\$470,532.46	\$61,618.69
2017	\$609,205.86	\$81,795.27
2018	\$733,215.19	\$93,439.77

Revenue increased from \$484.25K in 2015 to \$733.22K in 2018.

Profit increased from \$49.54K to \$93.44K over the same period.

An important observation is that 2016 revenue was slightly lower than 2015, while profit was higher.

KEY OBSERVATION

This demonstrates that revenue and profit do not necessarily move together in every period.

15. Customer Analysis

Customer performance was evaluated using both revenue and profit.

The analysis included:

- Top customers by revenue
- Top customers by profit
- Comparison of customer revenue and profitability

The results showed that customer revenue rankings and profit rankings do not perfectly align.

This demonstrates that customer sales volume alone does not fully represent customer value. A customer generating high revenue may not necessarily generate the highest profit.

Therefore, customer evaluation can benefit from considering both:

Revenue + Profitability

rather than revenue alone.

16. Product Analysis

Product-level analysis was used to identify high-performing products and understand product-level revenue contribution.

The Power BI dashboard includes a Top 10 Products by Revenue analysis. This provides a more detailed view of product performance beyond category and sub-category analysis.

Product analysis can help identify:

- High-revenue products
- Important product contributors
- Product concentration
- Areas requiring further profitability investigation

17. Python Analysis

Python was used after the SQL stage to perform data validation, exploratory data analysis, and business-focused analysis.

The Python workflow consisted of three notebooks:

Notebook	Purpose
01_data_cleaning.ipynb	Data Cleaning
02_eda.ipynb	Exploratory Data Analysis
03_business_analysis.ipynb	Business Analysis

Data Cleaning

Pandas was used to validate:

Dataset dimensions	Data types	Missing values
Duplicate records	Numerical summaries	Date fields
Sales	Discounts	Profit

Exploratory Data Analysis

The EDA examined:

Revenue distribution	Profit distribution	Category performance
Sub-category performance	Regional performance	Yearly trends
Discount patterns	Customer performance	Product performance

Business Analysis

Python was also used to calculate and investigate business metrics and patterns that supported the findings presented in the Power BI dashboard and final business analysis.

18. Power BI Dashboard

Power BI was used to create an interactive three-page business analytics dashboard. The dashboard combines KPI cards, analytical charts, customer analysis, regional analysis, product analysis, and interactive filters.

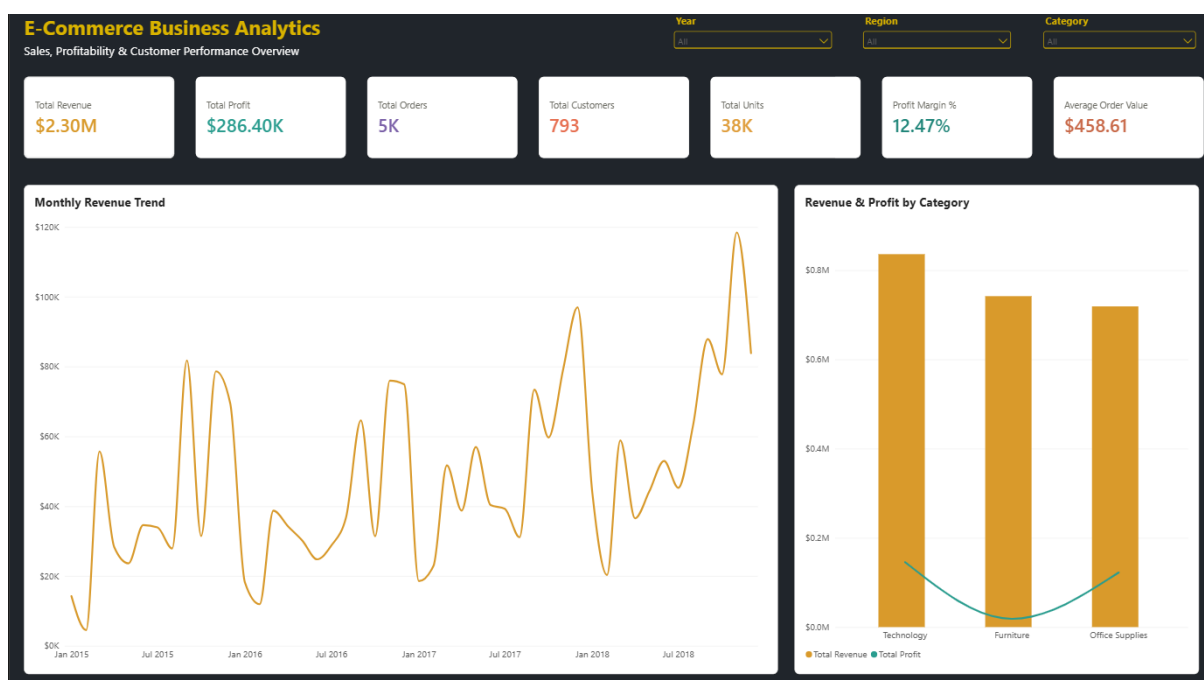
18.1 Executive Overview

The Executive Overview page provides a high-level summary of business performance. The page includes:

Total Revenue	Total Profit	Total Orders
Total Customers	Total Units	Profit Margin
Average Order Value	Monthly Revenue Trend	Revenue and Profit by Category

This page provides a starting point for understanding overall business performance.

Figure 1: Power BI Dashboard – Executive Overview



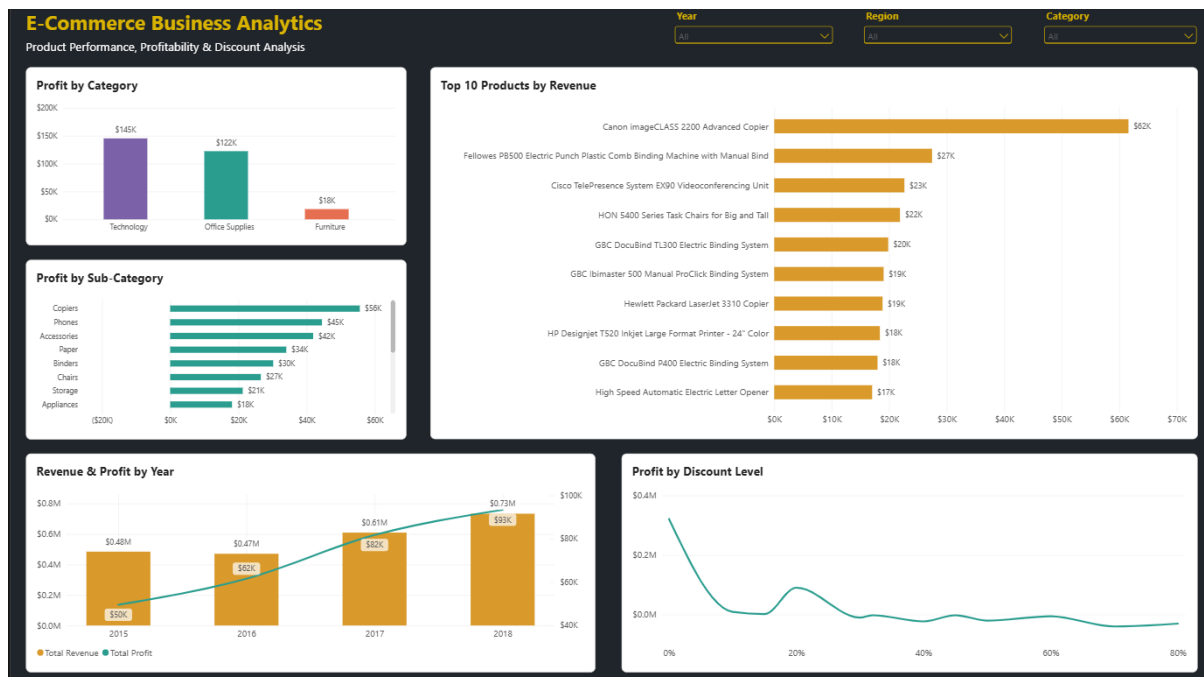
18.2 Product & Profitability

The Product & Profitability page focuses on product and profitability analysis. The page includes:

Profit by Category	Top 10 Products by Revenue	Profit by Sub-Category
Revenue and Profit by Year	Profit by Discount Level	

This page allows users to examine which product areas generate strong or weak profitability and how profitability changes across discount levels.

Figure 2: Power BI Dashboard – Product & Profitability



18.3 Customer & Regional Analysis

The Customer & Regional Analysis page focuses on customers and geographic performance. The page includes:

Revenue by Region	Revenue by Segment	Top 10 Customers by Revenue
Top 10 Customers by Profit	Revenue by State	

This page provides a more detailed view of customer and regional performance.

Figure 3: Power BI Dashboard – Customer & Regional Analytics



19. Interactive Dashboard Features

The Power BI dashboard includes synchronized filters across all three pages. The available filters are:

Year	Region	Category
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These filters allow users to dynamically explore the dashboard. For example, selecting a specific year, region, or category updates the relevant visualizations across the dashboard pages. This makes the dashboard more useful for interactive business analysis than a static report.

20. DAX Measures

Several DAX measures were created to calculate the key business KPIs.

Measure	DAX Formula
Total Revenue	Total Revenue = SUM(Sales[Sales])
Total Profit	Total Profit = SUM(Sales[Profit])
Total Orders	Total Orders = DISTINCTCOUNT(Sales[Order ID])
Total Customers	Total Customers = DISTINCTCOUNT(Sales[Customer ID])
Total Units	Total Units = SUM(Sales[Quantity])
Profit Margin	Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)
Average Order Value	Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)

These measures were used to power the KPI cards and analytical dashboard.

21. Key Business Insights

The complete analysis produced the following key findings.

01	Overall Business Performance The business generated: • \$2.30M revenue • \$286.40K profit • 12.47% profit margin • 5,009 orders This provides the overall baseline for evaluating business performance.
02	Technology and Office Supplies Demonstrate Strong Margins Technology generated a 17.40% profit margin, while Office Supplies generated a 17.04% margin. Both were above the overall business margin of 12.47%.
03	Furniture Has a Large Revenue Contribution but Low Profitability Furniture generated approximately \$742K in revenue but achieved only a 2.49% profit margin. This represents a significant difference between revenue contribution and profitability.

04

Tables Are a Major Loss-Making Sub-Category

Tables generated approximately -\$17.73K in profit with a -8.56% profit margin.
This makes Tables an important area for further investigation.

05

Copiers and Paper Show Strong Profitability

Copiers generated \$55.62K profit with a 37.20% margin.
Paper generated \$34.05K profit with a 43.39% margin.
These sub-categories demonstrate significantly stronger profitability than the overall business.

06

Higher Discount Levels Show Negative Aggregated Profit in Several Cases

Aggregated profit was negative at the 30% discount level and at several higher discount levels.
This highlights the importance of monitoring the profitability impact of discount strategies.

07

Regional Profitability Varies

Profit margins ranged from 7.92% in Central to 14.94% in West.
This demonstrates meaningful regional variation in profitability.

08

Revenue and Profit Increased Over the Analysis Period

Revenue increased from \$484.25K in 2015 to \$733.22K in 2018.
Profit increased from \$49.54K to \$93.44K.
This indicates overall growth across the analyzed period.

09

Customer Revenue Does Not Fully Represent Customer Profitability

Customer revenue rankings and profit rankings do not perfectly align.
This demonstrates why customer evaluation should consider profitability alongside revenue.

10

Revenue and Profit Can Move Differently

In 2016, revenue was slightly lower than in 2015, while profit was higher.
This demonstrates that revenue growth alone does not always describe changes in business profitability.

22. Business Implications and Recommendations

Based on the analytical findings, several areas can be considered for further business investigation and improvement.

22.1 Review Loss-Making Product Areas

Pricing, costs, discounting, and product-level performance should be investigated for loss-making sub-categories, particularly Tables.

22.2 Monitor Discount Strategies

Higher discount levels should be evaluated carefully because several discount levels at and above 30% showed negative aggregated profitability in the analyzed dataset.

Discount decisions should consider both revenue generation and profitability.

22.3 Investigate Furniture Profitability

Furniture contributes substantial revenue but has a relatively low profit margin. Further analysis could examine:

- Product-level margins
- Discount patterns
- Pricing
- Costs
- Sub-category performance

to understand the factors behind the lower profitability.

22.4 Investigate Regional Differences

The Central region has a lower profit margin than the other regions. Further analysis could investigate differences in:

- Product mix
- Discount levels
- Customer segments
- Sales patterns
- Product profitability

22.5 Study High-Margin Product Areas

Sub-categories such as Copiers and Paper demonstrate strong profitability. Their performance can be analyzed further to understand which factors contribute to their margins.

22.6 Evaluate Customers Using Multiple Metrics

Customer performance should be evaluated using both revenue and profit rather than relying only on sales volume.

22.7 Monitor Sustainable Growth

Revenue and profit should continue to be monitored together to determine whether future growth is accompanied by sustainable profitability.

23. Project Structure

The project repository is organized as follows:

```
E-Commerce-Business-Analytics/  
├── data/  
│   ├── cleaned/  
│   │   ├── ecommerce_cleaned_mysql.csv  
│   │   └── ecommerce_cleaned.xlsx  
│   └── raw/  
│       └── ecommerce_raw.csv  
├── excel/  
│   └── ecommerce_data_cleaning.xlsx  
├── powerbi/  
│   └── E-Commerce-Business-Analytics.pbix  
├── python/  
│   ├── 01_data_cleaning.ipynb  
│   ├── 02_eda.ipynb  
│   └── 03_business_analysis.ipynb  
├── sql/  
│   ├── 03_insert_data.py  
│   └── 04_business_analysis.sql  
├── screenshots/  
│   ├── executive-overview.png  
│   ├── product-profitability.png  
│   └── customer-regional-analysis.png  
└── README.md
```

24. Project Deliverables

The completed project includes:

No.	Deliverable
1	Cleaned transactional dataset
2	Excel data-cleaning file
3	MySQL database analysis
4	16 SQL business queries
5	Python data-cleaning notebook
6	Python EDA notebook
7	Python business-analysis notebook
8	Power BI dashboard
9	DAX KPI measures
10	Interactive dashboard filters
11	Business insights
12	Business recommendations
13	GitHub README documentation
14	Dashboard preview screenshots
15	Final project report

25. Conclusion

The E-Commerce Business Analytics project demonstrates a complete end-to-end data analytics workflow using Excel, MySQL, Python, and Power BI.

Starting from raw transactional data, the project progressed through data cleaning and validation, SQL-based business analysis, Python exploratory analysis, and interactive Power BI dashboard development.

The analysis identified important differences across categories, sub-categories, regions, customers, products, years, and discount levels.

KEY FINDINGS

- \$2.30M in total revenue
- \$286.40K in total profit
- 12.47% overall profit margin
- Strong profitability in Technology and Office Supplies
- Low profitability in Furniture
- Significant losses in the Tables sub-category
- Strong margins in Copiers and Paper
- Regional differences in profitability
- Negative aggregated profit at several higher discount levels
- Growth in revenue and profit from 2015 to 2018
- Differences between customer revenue and customer profitability

The final Power BI dashboard converts these findings into an interactive reporting experience, allowing users to explore business performance through multiple analytical perspectives and synchronized filters.

Overall, the project demonstrates the process of transforming raw transactional data into structured analysis, meaningful business insights, and interactive decision-support reporting.

26. Skills Demonstrated

Through this project, the following technical and analytical skills were demonstrated:

Data Analytics		
Data cleaning	Data validation	Exploratory data analysis
Business analysis	KPI development	Trend analysis
Profitability analysis	Customer analysis	Product analysis
Regional analysis		
SQL		
Data querying	Aggregation	Grouping
Filtering	Business KPI calculations	Customer and product analysis
Regional analysis	Discount analysis	Time-based analysis
Python		
Python data analysis	Pandas	Matplotlib
Data cleaning	EDA	Business metric calculation
Data aggregation		
Power BI		
Dashboard development	Data visualization	DAX
KPI cards	Interactive slicers	Cross-page filtering
Business reporting	Dashboard design	
Business Analysis		
Identifying profitability patterns	Comparing revenue and profit	Identifying loss-making areas
Interpreting business trends	Translating analytical results into business recommendations	

27. Final Project Summary

Project	E-Commerce Business Analytics
Dataset	Sample Superstore
Records	9,994
Revenue	\$2,297,201.07
Profit	\$286,397.79
Profit Margin	12.47%
Orders	5,009
Customers	793
Units Sold	37,873
Average Order Value	\$458.61
SQL Queries	16
Power BI Pages	3
Interactive Filters	Year, Region, Category
Primary Tools	Excel, MySQL, Python, Pandas, Matplotlib, Power BI, DAX

Project Workflow

